

A REFLECTION ON THE ELECTRON

Part II: The Dynamic Analysis

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Abstract

As a supplement to our previous work, this paper reconstructs the electron's dynamics as a spatiotemporal topological soliton. We derive the Dirac equation and Zitterbewegung effect from the coupling of massless chiral states and demonstrate that quantum measurement—specifically wave function collapse—is a deterministic nonlinear bifurcation in phase space. This framework establishes quantum mechanics as a macroscopic linear emergence of an underlying, deterministic nonlinear dynamical system.

1 Quantum Description of Electron Structure: Coupling of Mirror States and Emergence of Mass/Spin

Having established the geometric picture of the electron as a “topologically self-trapped photon pair,” in this section, we will reconstruct this underlying physical process entirely within the axiomatic framework of quantum mechanics (Hilbert space, state vectors, self-adjoint operators, and unitary evolution). We will demonstrate that the rest mass and half-integer spin of the electron are **emergent eigenstates** of massless bosonic basis vectors under specific off-diagonal Hamiltonian coupling.

1.1 Hilbert Space Truncation and the Construction of Spinor Basis

Under the first axiom of quantum mechanics, the pure state of a system is represented by a ray in a Hilbert space $\mathcal{H}$. For this system, we define a subspace restricted to longitudinal evolution (temporarily constraining transverse degrees of freedom) and choose the **Helicity-Momentum Basis** $\{|+\rangle, |-\rangle\}$:

- $|+\rangle$: A right-handed photon state with a momentum eigenvalue of $+p_z$ and a helicity (chirality) of $+1$;

- $|-\rangle$: A left-handed photon state with a momentum eigenvalue of $-p_z$ and a helicity (chirality) of -1 .

Under this basis, an arbitrary state $|\Psi\rangle$ of the electron can be represented by a two-component Dirac spinor spanned by a complex algebra:

$$|\Psi(z, t)\rangle = \psi_+(z, t)|+\rangle + \psi_-(z, t)|-\rangle \doteq \begin{pmatrix} \psi_+(z, t) \\ \psi_-(z, t) \end{pmatrix} \quad (1)$$

1.2 Unitary Evolution and Interaction Hamiltonian

According to the quantum mechanical Schrödinger equation $i\hbar\frac{\partial}{\partial t}|\Psi\rangle = \hat{H}|\Psi\rangle$, we must determine the total Hamiltonian operator $\hat{H}$ of the system. In an interaction-free vacuum (the linear limit), these two chiral states are mutually non-orthogonal and completely decoupled. Their free evolution is governed by the massless Weyl Hamiltonian $\hat{H}_0$:

$$\hat{H}_0 = -i\hbar\sigma_z\partial_z = \begin{pmatrix} -i\hbar\partial_z & 0 \\ 0 & i\hbar\partial_z \end{pmatrix} \quad (2)$$

In this state, the probability current diverges toward both ends at the speed of light, and the system's energy eigenspectrum is continuous, making it impossible to define a “rest particle.”

According to the “polarization-phase synchronization” topological locking condition derived previously, the non-linear response of the underlying grid is quantum-mechanically equivalent to introducing an **off-diagonal self-adjoint coupling operator** $\hat{H}_{int}$. This operator breaks the original chiral symmetry and induces transitional mapping between the states $|+\rangle$ and $|-\rangle$:

$$\hat{H}_{int} = mc^2\sigma_x = \begin{pmatrix} 0 & mc^2 \\ mc^2 & 0 \end{pmatrix} \quad (3)$$

Where the constant mc^2 represents the coupling matrix element for mutual conversion (corresponding to the Bragg reflection intensity in the classical picture). The total Hamiltonian is thus strictly equivalent to the $(1+1)$ D Dirac Hamiltonian:

$$\hat{H}_{total} = \hat{H}_0 + \hat{H}_{int} = -i\hbar\sigma_z\partial_z + mc^2\sigma_x \quad (4)$$

1.3 Non-commutativity of Velocity Operator and Spatial Localization

To explain why the electron is “at rest” (localized) within the quantum framework, we switch to the Heisenberg picture to examine the velocity operator $\hat{v}_z = \frac{d\hat{z}}{dt}$.

According to the Heisenberg equation of motion:

$$\hat{v}_z = \frac{1}{i\hbar}[\hat{z}, \hat{H}_{total}] = c\sigma_z = \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix} \quad (5)$$

This indicates that the instantaneous velocity eigenvalues of the electron are always $\pm c$ (the speed of light), which perfectly self-consists with the photon ontology. However, because the velocity operator $\hat{v}_z$ (containing σ_z) and the total Hamiltonian $\hat{H}_{total}$ (containing σ_x) **do not commute** ($[\sigma_z, \sigma_x] = 2i\sigma_y \neq 0$), the velocity $\hat{v}_z$ is no longer a conserved quantity.

Taking its second time derivative yields:

$$\frac{d\hat{v}_z}{dt} = \frac{1}{i\hbar}[\hat{v}_z, \hat{H}_{total}] = \frac{2mc^2}{\hbar}(c\sigma_y) \quad (6)$$

This is precisely the operator origin of the famous **Zitterbewegung (jitter effect)** in quantum mechanics. Physically, the off-diagonal term $mc^2\sigma_x$ causes extremely high-frequency quantum Rabi oscillations between $|+\rangle$ and $|-\rangle$ (at a frequency $\omega = 2mc^2/\hbar$). The moment the forward probability propagates a Compton wavelength at the speed of light, a coherent population inversion occurs, instantly converting it into backward propagation.

Quantum Conclusion: The expectation value of the macroscopic center-of-mass velocity $\langle\Psi|\hat{v}_z|\Psi\rangle$ strictly approaches 0 upon time averaging. The originally divergent massless probability wave achieves **the freezing and localization of probability flow** in space through the quantum coherent interference of off-diagonal interactions. The eigenenergy mc^2 required to maintain this high-frequency intrinsic interference is thus established as the macroscopic **rest mass** of the system.

1.4 Topological Projection and Emergence of SU(2) Spinor Symmetry

The greatest quantum statistical paradox faced under this framework is: how does a system composed of underlying bosonic fields (photon momentum flow) give rise to the 1/2 spin of a fermion?

This theory resolves this issue through the mechanism of **“fermionization of topological solitons”** in non-linear topological field theory. In the biorthogonal standing wave structure with polarization-phase locking, the topological morphology of the system is no longer a simple linear tensor product expansion, but rather forms a topological kink (Topological Soliton/Defect) containing a non-trivial topological charge (similar to a Hopf invariant) in three-dimensional space.

Under this strong non-linear topological constraint, the collective excited state space of the global system is forced into a **topological dimensional reduction**

constraint. The rigid locking of internal phases causes the state vector, when acted upon by external spatial rotation operators, to exhibit the geometric characteristics of a Dirac Belt or a scissor model: a spatial rotation of 2π introduces a geometric phase factor of $e^{-i\pi} = -1$ onto the internal spatiotemporal manifold. It requires a rotation of 4π to simultaneously untangle and restore the topological connection between the standing wave envelope and the internal polarization carrier wave.

Quantum Conclusion: The mass coupling term not only traps momentum but, through its self-induced topological configuration, forcibly maps (emerges) the underlying $SO(3)$ bosonic fluid algebra into the spinor space of the fundamental representation group $SU(2)$. The $1/2$ spin of the electron is not an intrinsic label of an elementary particle, but rather an inevitable geometric consequence that the macroscopic coherent evolution of the underlying bosonic field must obey when forming specific non-trivial topological solitons.

1.5 Summary

Through rigorous quantum mechanical operator formulations, we have proven that: **Mass is not an innate axiomatic property of a particle, but a dynamical eigenvalue generated by the coupling of massless eigenstates via off-diagonal operators; spin $1/2$ is also not an intrinsic label of elementary particles, but a topological group representation that the coupled-state wave function must obey to maintain coherent evolution on the spatiotemporal manifold.** The model in this paper is not opposed to the first principles of quantum mechanics; rather, it provides an underlying physical ontological picture for the various abstract mathematical symbols (spinors, mass terms, matrix algebra) in the Dirac equation.

2 The Dynamical Essence of Forces and Nonlinear Collapse of Quantum Measurement

In the preceding sections, we established the static and spin structures of the free electron as a four-dimensional spatiotemporal topological soliton (biorthogonal phase-locked standing wave). This section will extend the theory to the dynamic interactions between the soliton and external environmental fields. We will prove that classical electromagnetic force (Lorentz force) and minimal coupling in quantum mechanics are perfectly isomorphic in terms of underlying fluid geometry. Furthermore, we will thoroughly deconstruct the mechanical physical origin of “wave function collapse” and “spin quantization” in the Stern-Gerlach experiment from

a purely classical deterministic perspective, utilizing non-linear dynamical phase space attractors.

2.1 Soliton Kinematics: De Broglie Coherent Envelope and Interaction Energy

Under the framework of “Computational Realism,” the electron entity is a self-consistent four-dimensional potential field soliton A_s^μ . When the soliton translates as a whole across the underlying grid at a macroscopic velocity $\mathbf{v}$, due to the finiteness of the information propagation speed c between grid nodes, various points inside the soliton will experience a **directional information update latency** when participating in recursive synchronization. To maintain the topological closure and global synchronization of the standing wave structure, the grid nodes at the front and rear of the soliton must adjust their local oscillation phases to compensate for the information feedback lag caused by displacement. This **spatiotemporal synchronization compensation**, based on the grid’s recursive mechanism, causes the original high-frequency rotational phase inside the soliton to accumulate a linear phase along the direction of motion, thereby giving rise to a **De Broglie Coherent Envelope**:

$$A_s^\mu(\mathbf{x}, t) \approx A_{static}^\mu(\mathbf{x} - \mathbf{v}t) e^{i(\mathbf{k}_{dB} \cdot \mathbf{x} - \omega t)} \quad (7)$$

At this point, the soliton and the external background field A_{ext}^μ interfere through the scalar product of their four-potentials, constituting an additional **Interaction Energy Density** for the system:

$$u_{int} = 2\epsilon_0\omega^2 A_s^\mu A_{\mu, ext} \quad (8)$$

From the perspective of relativistic wave dynamics, the De Broglie wave is not an independent objective entity, but the projection of the particle’s intrinsic frequency $\omega_0 = m_0c^2/\hbar$ onto the observer’s spatiotemporal coordinates. The complex phase $S = \int p_\mu dx^\mu$, defined by the action integral, physically represents the phase-alignment envelope of the particle’s internal clock. The phase gradient ∇S of this envelope directly determines the superposition interference pattern of the interaction energy in space.

2.2 Fluid Topological Isomorphism between Classical Electromagnetic Force and Quantum Minimal Coupling

The so-called “force” in macroscopic mechanics is essentially a **macroscopic fluid dynamical emergence** driven by the underlying grid to eliminate the local phase

space imbalance caused by the interaction energy u_{int} , forcing the soliton's center of mass to continuously move toward paths with lower phase potential energy.

2.2.1 Electric Force: Scalar Pressure Relaxation Thrust and Wavefront Expansion/Contraction

The external electric field is composed of a scalar potential A_{ext}^0 . Here, the interaction energy term manifests as the coupling between the soliton's De Broglie envelope and the background scalar pressure: $u_{int} = 2\epsilon_0\omega^2 A_s^0 A_{ext}^0$.

- **Grid Fluid Picture:** The external potential gradient ∇A_{ext}^0 breaks the energy density balance within the De Broglie envelope, creating a slight difference in the vacuum grid saturation pressure between the front and rear of the soliton. This pure scalar pressure difference acts on the coherent envelope, generating a “relaxation thrust” directed toward the low-pressure region. Its macroscopic resultant force is equivalent to the variation of the interaction energy with respect to the center-of-mass coordinates: $\mathbf{F}_E = -\frac{\partial}{\partial \mathbf{x}_0} \int u_{int} d^3x = q\mathbf{E}$.
- **Phase Dynamics Isomorphism:** In quantum mechanics, this is equivalent to the minimal coupling rule $\partial_t \rightarrow \partial_t + i\frac{q}{\hbar}A_{ext}^0$. The potential gradient alters the local spatial phase evolution frequency (the rate of time flow), forcing the wavefront of the De Broglie wave to compress or stretch in space. To maintain the phase continuity of the wavefront, the macroscopic momentum must evolve over time, and the force emerges as the explicit time derivative of the action phase gradient: $\mathbf{F}_E = \frac{d}{dt}(\nabla S) = -\nabla(qA_{ext}^0)$.

2.2.2 Magnetic Force: Topological Magnus Effect and Waveguide Deflection

The external magnetic field is composed of a vector potential $\mathbf{A}_{ext}$, and the interaction energy term becomes $u_{int} = -2\epsilon_0\omega^2 \mathbf{A}_s \cdot \mathbf{A}_{ext}$.

- **Grid Fluid Picture:** Due to motion, the De Broglie envelope already contains translational vector information, which participates in the coupling alongside the soliton's internal spin chiral vortex potential A_s^{vortex} . Under this asymmetric superposition, the momentum flow on one side of the soliton vortex ring is strengthened, while the other side is canceled out, leading to a deviation in grid saturation across the transverse plane. This **Topological Magnus Effect** in fluid mechanics forces the entire standing wave structure to deflect. The convective derivative directly yields the Lorentz force form: $\mathbf{F}_B = \int (\mathbf{J}_s \times \nabla \times \mathbf{A}_{ext}) d^3x = q(\mathbf{v} \times \mathbf{B})$.

- **Phase Dynamics Isomorphism:** In quantum mechanics, this manifests as the minimal coupling of the Aharonov-Bohm (A-B) effect $\nabla \rightarrow \nabla - i\frac{q}{\hbar}\mathbf{A}_{ext}$. The background vector potential forcibly superimposes a chiral bias onto the moving phase gradient, altering the equiphase surface curvature of the wavefront in transverse space, causing the particle’s waveguide direction to be topologically steered during movement: $\mathbf{F}_B = q\mathbf{v} \times (\nabla \times \mathbf{A}_{ext})$.

2.3 Dynamical Reduction of the Stern-Gerlach Experiment: Coherence Breaking and Nonlinear Collapse

In the mainstream Copenhagen interpretation, the discrete splitting of electron spin in the Stern-Gerlach (S-G) experiment is attributed to a “measurement-induced wave function collapse” that transcends classical physics. This section will discard any quantization postulates. Relying solely on “standing wave coherence” and “non-linear phase space attractors” from non-linear wave dynamics, we will deconstruct the pure fluid topological origin of “wave function collapse” through a dual-picture comparison.

2.3.1 Environmental Coupling and Phase Mismatch in Gradient Fields

When an electron with a continuous, random initial spin angle α enters the extreme gradient magnetic field ∇B_z of the S-G experiment, spatial translational symmetry is broken, and the particle’s internal state begins to couple strongly with the asymmetric external environment.

- **Grid Fluid Picture (Topological Phase Mismatch):** In the underlying grid, the electron is a topological standing wave vortex with a transverse scale of λ_c (Compton wavelength). The two sides of the vortex cross-section experience objective differences in the background vector potential. Due to the Aharonov-Bohm (A-B) effect, this asymmetric environment forcibly introduces a **Differential Phase Shift** inside the standing wave cavity. Geometric projection shows that this phase mismatch is proportional to the effective span of the vortex ring along the gradient direction: $\Delta\Phi \propto |\nabla B_z|\lambda_c \sin \alpha$.
- **Quantum Physics Isomorphism (Spin-Orbit Entanglement):** In quantum mechanics, this is equivalent to substituting the Hamiltonian $H = -\boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{z})$ into the Schrödinger equation. Because the magnetic field is a function of the spatial coordinate z , the particle’s internal spin degrees of freedom become inseparably entangled with its external spatial translational degrees of freedom, causing the wave function to begin dispersing in phase space.

2.3.2 Loss of Standing Wave Stability and Vulnerability of Intermediate States

Any non-parallel polarization state ($\alpha \neq 0, \pi$) faces a breakdown of coherence in the gradient field.

- **Grid Fluid Picture (Momentum Flow Escape):** The prerequisite for the electron's stable existence is the perfect interference closure (Bragg reflection) of the forward and backward waves. In wave interference theory, the resonance overlap efficiency caused by a phase mismatch is proportional to $\cos(\Delta\Phi) \approx 1 - \frac{1}{2}(\Delta\Phi)^2$. This implies that the **loss of stability (i.e., the rate Γ_{escape} at which internal momentum flow escapes outward)** due to phase misalignment is strictly proportional to the square of the phase difference:

$$\Gamma_{escape}(\alpha) \propto (\Delta\Phi)^2 = \beta |\nabla B_z|^2 \sin^2 \alpha \quad (9)$$

This demonstrates that all tilted intermediate states will cause the standing wave cavity to “leak,” posing a threat of disintegration to the momentum flow.

- **Quantum Physics Isomorphism (Uncertainty of Superposition States):** In quantum mechanics, this manifests as the intrinsic uncertainty of a superposition state. A pure state at an arbitrary angle α , $|\psi\rangle = \cos(\alpha/2)|\uparrow\rangle + \sin(\alpha/2)|\downarrow\rangle$, has a variance (uncertainty) under the measurement operator S_z that is exactly proportional to $\sin^2 \alpha$. The statistical properties described by “probability variance” in QM correspond to the “actual physical rate of momentum flow escape” in the underlying fluid.

2.3.3 Nonlinear Bifurcation and Deterministic Collapse of the Wave Function

To prevent the disintegration of the momentum flow, the system's internal non-linear feedback mechanism is activated, forcing a topological reorganization of the phase space.

- **Grid Fluid Picture (Bistable Attractor Roll-down):** The escaping momentum flow generates a self-feedback correction on the soliton body, driving the principal spin axis α to spontaneously evolve toward a state that **minimizes the escape rate**. Its dynamical equation is $\frac{d\alpha}{dt} \propto -\frac{\partial \Gamma_{escape}}{\partial \alpha}$. Differentiating $\sin^2 \alpha$ derives the non-linear bistable bifurcation equation:

$$\frac{d\alpha}{dt} = -\lambda \sin(2\alpha) \quad (10)$$

(where λ is a reorganization rate correlated with the gradient field strength). Within an extremely short relaxation time, the massive $-\lambda \sin(2\alpha)$ feedback makes $\alpha = \pi/2$ a highly unstable saddle point. Any intermediate state deviating from the absolute polar axes will undergo an **Avalanche-like Roll-down**, being forcibly twisted and plummeting into one of two **Stable Attractors** ($\alpha = 0$ or $\alpha = \pi$) where no momentum escape occurs.

- **Quantum Physics Isomorphism (Measurement Collapse Axiom):** This is precisely the “**Wave Function Collapse**” or “von Neumann projection” in quantum mechanics.

2.3.4 Macroscopic Trajectory Splitting and Emergence of Quantization

After completing the collapse and reorganization on a microscopic time scale, the system enters the macroscopic observation phase.

- **Grid Fluid Picture (Macroscopic Lorentz Force Diversion):** Once all continuously distributed solitons have instantly completed their “non-linear bifurcation archiving” and entirely differentiated into the two topological steady states of absolutely upward ($\alpha = 0$) or absolutely downward ($\alpha = \pi$), they begin to be governed purely by the macroscopic translational force $\mathbf{F}_z = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$. Because $\boldsymbol{\mu}$ now possesses only two discrete absolute directions, the particle flow forms two sharply fractured trace lines on the macroscopic screen.
- **Quantum Physics Isomorphism (Discretization of Eigenvalues):** The experiment ultimately observes the phenomenon of “spatial quantization” of spin—measurement results can only yield discrete eigenvalues of $\pm\hbar/2$, with no transitional intermediate states existing.

Summary of this Section: Through a one-to-one mapping between the grid fluid picture and the quantum picture, we have proven that: “quantum measurement collapse” and “discrete quantization” are **spontaneous non-linear bistable bifurcations and deterministic topological phase transitions initiated by underlying finite-volume continuous standing wave fluids when subjected to extreme constraints in order to maintain interference stability**. Quantum mechanics is merely a macroscopic linear statistical summary of this ultra-fast, underlying non-linear dynamical process. **The quantum discreteness of spin is not an inexplicable a priori axiom, but rather the topological symmetry breaking and macroscopic emergence that must occur in the phase space attractors of underlying continuous fluid matter waves when encountering extreme boundary conditions.**

3 Hydrodynamic Analysis and Geometric Decoding of the Fine-Structure Constant

This section will demonstrate, by analyzing the four-dimensional spatiotemporal partial differential equations of envelope evolution, that **the fine-structure constant α is fundamentally a geometric and kinematic ratio**—specifically, the “Envelope Strain Parameter” between the macro-gradient of the slowly varying envelope and the spatial frequency of the micro-carrier wave propagating at the speed of light, following the spatial relaxation of the envelope within an inhomogeneous external nuclear potential field.

3.1 The Slowly Varying Envelope Approximation (SVEA) and the Definition of the “Envelope Strain Parameter”

When deriving the Dirac equation in Sections 12.1 and 12.3, the reduction of the higher-order four-dimensional D’Alembertian operator ($\square$) to first-order equations necessitated the introduction of the **Slowly Varying Envelope Approximation (SVEA)**, a cornerstone of non-linear wave physics.

Within this spatiotemporal soliton (the electron), the wave field is partitioned into two physical structures operating across vastly different scales:

1. **Microscopic Carrier Wave:** Characterized by a wavenumber $k_c = \frac{mc}{\hbar} = \frac{1}{\lambda_c}$ (the inverse of the reduced Compton wavelength), this wave establishes standing waves across the grid and propagates phase variations at the speed of light c .
2. **Macroscopic Envelope $F(r)$:** Represents the macroscopic physical distribution and orbital morphology of the electron fluid in space.

The SVEA mathematically requires the spatial variations of the envelope to be sufficiently smooth. To quantitatively measure this geometric feature, we define a dimensionless “**Envelope Strain Parameter η** ”, representing the ratio of the spatial gradient of the envelope to the spatial frequency of the carrier wave:

$$\eta \equiv \frac{|\nabla F|}{k_c |F|} \quad (11)$$

This parameter η carries a highly explicit physical meaning: **it measures the severity of the spatial decay and variation of the macroscopic fluid envelope relative to the physical scale of the underlying Compton grating.**

3.2 Reconceptualizing the Morphology of the Free Electron

We calculate the envelope scale of the electron soliton in free space (in the absence of an external nuclear potential). According to the **time-dependent coupled-mode equations** derived in Section 12.5.2, for a free electron in its rest frame, the macroscopic envelope is time-independent ($\partial_t F = \partial_t B = 0$), and its pure spatial evolution equations satisfy:

$$i \left(c \frac{\partial F}{\partial z} \right) = (c\kappa)B \quad (12)$$

$$i \left(-c \frac{\partial B}{\partial z} \right) = (c\kappa)F \quad (13)$$

Differentiating the first equation with respect to space and substituting it into the second yields the second-order spatial evolution equation for the forward envelope F :

$$\frac{\partial^2 F}{\partial z^2} = \kappa^2 F \quad (14)$$

The stable physical trial solution to this equation takes the form of an exponential decay: $F(z) \propto e^{-\kappa|z|}$. Thus, the characteristic decay length (physical radius) of the free-space electron envelope is uniquely determined by the coupling coefficient κ :

$$L_{free} = \frac{1}{\kappa} \quad (15)$$

Invoking the physical constant anchoring relation $c\kappa \equiv \frac{mc^2}{\hbar}$, we solve for the characteristic envelope scale of the free electron:

$$L_{free} = \frac{\hbar}{mc} \equiv \lambda_c \quad (16)$$

Substituting L_{free} back into the definition of the strain parameter yields the envelope strain of the free electron soliton:

$$\eta_{free} = \frac{1/L_{free}}{k_c} = \frac{1/\lambda_c}{1/\lambda_c} = 1 \quad (17)$$

Thus, the free electron resides in a rigid limit state of its own non-linear self-focusing. Its envelope gradient is perfectly balanced with the microscopic carrier frequency ($\eta = 1$), causing it to manifest in space as a highly compact “hard-core” soliton with a physical radius equal to the reduced Compton wavelength λ_c .

3.3 Envelope Relaxation and the Adaptive Redshift Mechanism in Non-Uniform Environments

When the electron enters the non-uniform scalar pressure field of a nucleus, $V_N(r) = -\frac{e^2}{4\pi\epsilon_0 r}$, the space it occupies is no longer symmetric or homogeneous. Permeated

by the nuclear potential field, the electron wave packet undergoes spatial relaxation and expands, forming a three-dimensional macroscopic envelope $F(r) \propto e^{-r/a_0}$ with a radius a_0 .

At this juncture, the steady-state self-consistency of the wave packet must be governed by the four-dimensional covariant spacetime operator. We examine the dual algebraic alignment of this system across both temporal and spatial components:

$$\left[\frac{1}{c^2} \left(\partial_t + i \frac{q}{\hbar} V_N(r) \right)^2 - \nabla^2 + \frac{1}{a^2} \right] \Psi = 0 \quad (18)$$

3.3.1 The “Hanging Constant Term” Instability

Assume that after expansion, the clock frequency of the electron’s internal carrier wave still retains its free-state intrinsic frequency $\omega_0 = mc^2/\hbar$, such that its spatiotemporal wave function is written as $\Psi = \phi(r)e^{-i\omega_0 t}$.

Substituting this wave function into the covariant wave equation, expanding the temporal terms, and canceling the common temporal phase factor $e^{-i\omega_0 t}$ reduces the equation to:

$$\left(-\frac{m^2 c^2}{\hbar^2} + \frac{2mq}{\hbar^2} V_N(r) \right) \phi - \nabla^2 \phi + \frac{1}{a^2} \phi = 0 \quad (19)$$

Since the free-grid constant anchoring relation dictates $\frac{1}{a^2} \equiv \frac{m^2 c^2}{\hbar^2}$, these two terms cancel. Substituting the charge $q = -e$ and the potential field $V_N(r)$ simplifies the equation to:

$$\nabla^2 \phi + \frac{2me^2}{4\pi\epsilon_0 \hbar^2 r} \phi = 0 \quad (20)$$

We substitute the exponential decay trial solution $\phi = \phi_0 e^{-r/a_0}$ (whose Laplacian in spherical coordinates is $\nabla^2 \phi = \left(\frac{1}{a_0^2} - \frac{2}{ra_0} \right) \phi$) into the equation above. Grouping like terms yields an algebraic balance equation with respect to the spatial coordinate r :

$$\underbrace{\frac{1}{a_0^2}}_{\text{Constant Term}} + \underbrace{\left(\frac{2me^2}{4\pi\epsilon_0 \hbar^2} - \frac{2}{a_0} \right) \frac{1}{r}}_{\text{Gradient Term}} = 0 \quad (21)$$

For this equation to hold strictly across all space, the constant term and the gradient term must each be independently aligned to zero:

- To align the gradient term, the spatial scale must be locked: $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2}$ (which defines the Bohr radius).
- However, once a_0 is fixed, the constant term $\frac{1}{a_0^2}$ is left “hanging,” constituting an unresolvable residual spatial curvature across all space.

The inability to align the constant term physically implies that the internal phase of the wave packet cannot synchronize across different locations in space. This superfluous residual curvature continuously generates phase-space leakage, meaning that the expanded envelope is dynamically unstable. It would rapidly dissipate its phase outward, failing to establish any stationary, self-consistent standing wave in the non-uniform field.

3.3.2 Adaptive Dual Alignment of Space and Time

This dissipation process, analogous to the self-phase modulation (SPM) mechanism of phase velocity in non-linear optics, constructively utilizes the soliton's internal intrinsic oscillation frequency as a degree of freedom for **adaptive self-tuning**. As energy leaks, a redshift spontaneously occurs within the system. Assuming a minute phase shift $\Delta\omega$, the intrinsic frequency is adjusted to:

$$\omega_{bound} = \omega_0 - \Delta\omega \quad (22)$$

Consequently, the second-order temporal derivative operator generates an additional constant compensation term. Substituting the new temporal phase factor $e^{-i\omega_{bound}t}$ into the covariant equation, the temporal expansion becomes:

$$-\frac{\omega_{bound}^2}{c^2} \approx -\frac{\omega_0^2}{c^2} + \frac{2\omega_0\Delta\omega}{c^2} = -\frac{1}{a^2} + \frac{2m\Delta\omega}{\hbar} \quad (23)$$

Reintroducing this self-tuned result back into the global spatial balance equation, the expression transforms into:

$$\underbrace{\left(\frac{1}{a_0^2} - \frac{2m\Delta\omega}{\hbar}\right)}_{\text{Constant Term}} + \underbrace{\left(\frac{2me^2}{4\pi\epsilon_0\hbar^2} - \frac{2}{a_0}\right)\frac{1}{r}}_{\text{Gradient Term}} = 0 \quad (24)$$

By introducing the self-tuned redshift $\Delta\omega$, **the once-hanging constant term acquires a perfect cancellation pathway.**

To achieve dynamical closure in both space and time, the system is geometrically forced to reach a dual equilibrium:

- **Gradient Alignment (Torque Balance Constraint):** To balance the spatial gradient torque and prevent the disassembly of the soliton, the scale must be locked as:

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} \quad (25)$$

- **Constant Alignment (Phase Synchronization Constraint):** To ensure that the internal clock phase remains synchronized throughout the expanded

envelope, the redshift $\Delta\omega$ must be precisely adjusted to completely cancel the residual spatial curvature $\frac{1}{a_0^2}$:

$$\Delta\omega = \frac{\hbar}{2ma_0^2} \quad (26)$$

At this point, a new equilibrium is established: **the only self-consistent, adaptive frequency adjustment made by the electron standing wave to preserve spatiotemporal closure in a non-uniform environment and prevent dissipative collapse results in a frequency redshift $\Delta\omega = \frac{\hbar}{2ma_0^2}$, thereby “releasing binding energy” ($E_b \equiv \hbar\Delta\omega$).**

3.4 Algebraic Closed Loop and the Structural Ontology of the Fine-Structure Constant

Now, we substitute the steady-state envelope decay length $L_{bound} = a_0$, obtained from the adaptive dual equilibrium, back into the system’s fundamental definition of the dimensionless “envelope strain parameter”:

$$\eta_{bound} = \frac{1/a_0}{k_c} = \frac{\lambda_c}{a_0} = \frac{\hbar}{mc} \cdot \left(\frac{me^2}{4\pi\epsilon_0\hbar^2} \right) \quad (27)$$

Through straightforward algebraic simplification, we obtain the identity:

$$\eta_{bound} = \frac{e^2}{4\pi\epsilon_0\hbar c} \equiv \alpha \quad (28)$$

This rigorous algebraic alignment completes a deeper physical closed loop: **it reveals the structural ontology of the fine-structure constant. The relative “strain parameter η_{bound} ” of the macroscopic envelope to the microscopic carrier wave, as defined in this system, is strictly equivalent in value, dimension, and physical meaning to the fine-structure constant α .**

3.5 Deep Physical Corollaries and Geometric Isomorphism of Relativistic Corrections

Equating the fine-structure constant to the “envelope strain parameter” ($\eta \equiv \alpha$) of the fluid soliton provides a foundational geometric explanation for existing relativistic quantum phenomena:

1. Justification of the First-Order Wave Equation (Dirac Equation):

Previously in Section 12.3, we derived the linearized first-order Dirac equation by dropping the second-order partial derivative terms (under the SVEA). The

fundamental justification for this approximation is that $\eta_{\text{bound}} = \alpha \approx 1/137 \ll 1$. This proves that when the electron fluid forms a macroscopic standing wave outside the nucleus, the distortion of its envelope is exceedingly minute (less than one percent of the underlying self-trapping limit). Because α is sufficiently small, the slowly varying envelope approximation (SVEA) holds to high accuracy, rendering the first-order quantum operators extremely precise.

2. Geometric Mapping of Higher-Order Terms and Relativistic Corrections:

Through the spatiotemporal differential equivalence of phase gradients, the spatial geometric strain η dynamically maps strictly to the ratio of the macroscopic phase velocity to the speed of light, $\beta = v/c$. When the envelope changes rapidly due to intense fields and second-order derivatives can no longer be ignored, the higher-order operator truncation must be expanded non-linearly, i.e., $\sqrt{1 - \eta^2} \approx 1 - \frac{1}{2}\eta^2$. Since η maps to v/c , this purely non-linear geometric truncation expansion of $\sqrt{1 - \eta^2}$ is macroscopically isomorphic to the Lorentz factor $\sqrt{1 - (v/c)^2}$ in relativistic kinematics. **This provides the fundamental reason why all relativistic fine-structure correction terms in atomic spectra (such as spin-orbit coupling and the Darwin term) always scale with α^2 .** This is not an ad hoc relativistic postulate, but the inevitable algebraic consequence of the higher-order non-linear spatial response of the standing wave envelope under high curvature.

3. Topological Limits of the Four-Dimensional Grid and Vacuum Decay:

On the ultimate physical horizon, α measures the “slippage and fault-tolerance limit” of the vacuum grid. When the external field strength becomes extremely large (such as near a superheavy nucleus with $Z \approx 137$), the envelope strain parameter approaches unity ($\eta \rightarrow 1$), the SVEA completely breaks down, and the envelope gradient converges to the microscopic Compton carrier wave scale. At this boundary, the physical limits of the grid can no longer sustain the synchronized closed loop of the internal wave packet, leading to the rupture of the fluid standing wave. In the underlying fluid picture, this intuitively corresponds to the breakdown of the Dirac vacuum and the spontaneous generation of electron-positron pairs when $Z > 137$ in relativistic quantum mechanics.

4 Bound-State Wave Interference and Impedance Matching: Spatial Scale Coordination and the Geometric Emergence of the Vacuum “Interference Funnel”

In the intersection of classical electrodynamics and quantum mechanics, a spatial scale gap of nearly four orders of magnitude exists between the physical dimension of the ground-state hydrogen atom ($a_0 \sim 0.05$ nm) and the electromagnetic wavelength associated with its transitions ($\lambda \sim 100$ nm). Traditional theories address this by introducing phenomenological probabilistic concepts such as “absorption cross-sections” and “transition probabilities.” This section will demonstrate that this mechanism can be explained via **dynamical operator cancellations in the covariant wave equation**, which spontaneously generate a macroscopic “**Interference Funnel**” in the vacuum surrounding the atom. This structure achieves deterministic energy guiding and impedance matching at both geometric and kinematic levels.

4.1 Dynamical Operator Decomposition and Frequency Definition in the Covariant Wave Equation

To precisely delineate the geometric role of each physical interaction term in spatial evolution, we return the wave function to the original stationary covariant wave equation:

$$\left[\nabla^2 - \frac{2m\Delta\omega}{\hbar} + \frac{2me^2}{4\pi\epsilon_0\hbar^2 r} \right] \phi(r) = 0 \quad (29)$$

Substituting the ground-state hydrogen trial solution $\phi = \phi_0 e^{-r/a_0}$ and utilizing the expansion of the spherical Laplacian $\nabla^2 \phi = \left(\frac{1}{a_0^2} - \frac{2}{ra_0} \right) \phi$, we can completely expand the operator balance of this PDE across all space as:

$$\underbrace{\frac{1}{a_0^2} \phi}_{\text{Term 1}} - \underbrace{\frac{2}{ra_0} \phi}_{\text{Term 2}} - \underbrace{\frac{2m\Delta\omega}{\hbar} \phi}_{\text{Term 3}} + \underbrace{\frac{2}{ra_0} \phi}_{\text{Term 4}} = 0 \quad (30)$$

Where:

Term 1 represents the Global Self-Tension;

Term 2 represents the Spherical Dispersion;

Term 3 represents the Clock Redshift Compensation;

Term 4 represents the Local Nuclear Constraint.

Analyzing the signs and boundary conditions of this dynamical operator structure reveals that, due to the immense asymmetric mass difference between the electron and the nucleus, the nuclear potential acts on the electron with near rigidity. Under the influence of the nuclear $1/r$ potential, the electron's four-dimensional potential distribution generates **a hierarchical, coordinated locking of two distinct modes of constraint across different spatial dimensions**:

1. **Local Gradient-Level Constraint (Coordination of Term 4 and Term 2, the ω_c mode)**: In the near-field, the nuclear potential term (Term 4: $+\frac{2}{ra_0}\phi$) enters with a positive sign. It strictly cancels the geometric spherical dispersion term (Term 2: $-\frac{2}{ra_0}\phi$) across all space. This indicates that **the nuclear potential dynamically counters the wave packet's tendency to disperse due to spherical geometry, thereby establishing a steady-state "gradient-level constraint" in the local gradient dimension.**
2. **Global Scale-Level Constraint (Coordination of Term 1 and Term 3, the ω_{rel} mode)**: In the far-field, the self-tension term resulting from the second spatial derivative of the exponentially decaying envelope (Term 1: $+\frac{1}{a_0^2}\phi$) represents the soliton's intrinsic **"global self-contraction constraint"** maintained by non-linear mechanisms. This term aligns strictly with the clock redshift term representing the self-tuned intrinsic frequency (Term 3: $-\frac{2m\Delta\omega}{\hbar}\phi$), thereby establishing a steady-state "scale-level constraint" on the global envelope dimension.

Based on the characteristic diffusion coefficient $D = \frac{\hbar}{2m}$, these two levels of coordinated constraint correspond to two independent characteristic constraint frequencies at the characteristic boundary $r = a_0$:

- **Local gradient-level constraint frequency (ω_c)**:

$$\omega_c = D \cdot \frac{2}{a_0^2} = \frac{\hbar}{ma_0^2} = \frac{mc^2\alpha^2}{\hbar} \quad (31)$$

- **Global scale-level constraint frequency (ω_{rel})**:

$$\omega_{rel} = D \cdot \frac{1}{a_0^2} = \frac{\hbar}{2ma_0^2} = \frac{1}{2} \frac{mc^2\alpha^2}{\hbar} \quad (32)$$

Here, the proportional relationship $\omega_c = 2\omega_{rel}$ directly yielded by the operator geometry is the pure geometric emergence of the **Virial Theorem** (i.e., $2\langle T \rangle = -\langle V \rangle$) within the stationary covariant wave equation. It demonstrates that the local constraint intensity is numerically exactly twice the global constraint intensity.

4.2 Envelope Beat Frequency and the Spatial Deployment of the Macroscopic “Interference Funnel”

In the space outside the nucleus occupied by the 1s orbital fluid, these two non-synchronized characteristic oscillations interfere. Their residual temporal curvature (the system’s adaptive redshift $\Delta\omega$) manifests as a macroscopic **intrinsic envelope beat frequency**:

$$\omega_{beat} = \omega_c - \omega_{rel} = \frac{1}{2} \frac{mc^2\alpha^2}{\hbar} \quad (33)$$

This high-frequency beat oscillation radiates outward into the vacuum at the effective speed of light c , carving large-scale interferometric Moiré patterns in the surrounding space. The wavelength of this macroscopic wave in the vacuum (which defines the characteristic spatial scale of the interference funnel) is given by:

$$\lambda_{funnel} = \frac{2\pi c}{\omega_{beat}} = \frac{4\pi\hbar}{mc\alpha^2} \quad (34)$$

Utilizing the Bohr radius identity $a_0 = \frac{\hbar}{m\alpha}$, this wavelength can be expressed in an exceptionally elegant, dimensionless geometric mapping form:

$$\lambda_{funnel} = \frac{4\pi}{\alpha} a_0 \quad (35)$$

Given that the dimensionless envelope strain parameter (the fine-structure constant) is $\alpha \approx 1/137.036$, we calculate:

$$\lambda_{funnel} \approx 1722a_0 \approx 91.13 \text{ nm} \quad (36)$$

This value is strictly consistent with the Lyman limit wavelength of the hydrogen atom within experimental precision. This algebraic identity proves that the ground-state hydrogen atom is not a rigid soliton confined within the Bohr radius; rather, through the intrinsic envelope beat mechanism, it deploys a geometric “virtual slot network” formed by interference fringes across hundreds of nanometers in the surrounding vacuum.

4.3 Parametric Resonance, Impedance Matching, and Non-Linear Self-Focusing

When electromagnetic radiation (a photon soliton with wavelength $\lambda \sim 100 \text{ nm}$) traverses this spatial region, the physical mechanism of its absorption cross-section can be explained by the following deterministic evolutionary stages:

1. **Parametric Resonance and Impedance Matching:** The characteristic spatial scale of the incident photon matches the geometric boundary of the

“interference funnel,” λ_{funnel} . At this stage, the funnel structure acts as a **graded impedance transformer** between the macroscopic vacuum photon and the microscopic electron soliton, locking phases and guiding the energy flow through parametric resonance.

2. **Centripetal Non-Linear Self-Focusing:** Once resonance is established, the system deviates from the limit of linear superposition, and the non-linear feedback of the underlying grid is activated. The vector momentum flow of the photon undergoes self-focusing collapse along the gradient pressure channels within the interference funnel, compressing the transverse wavefront energy and drawing it deep into the funnel’s core.
3. **Topological Reconstruction and Virtual Slot Solidification:** The converging momentum and energy streams feed into the higher-order stationary nodes (such as the spatial potential wells of the 2p or 3p orbitals) pre-carved in space by the 1s-nucleus beat frequency wave. Upon absorbing momentum, the 1s fluid undergoes topological reconstruction, achieving a deterministic transition to a higher energy state.

4.4 Extended Discussion: Large-Scale Interferometric Guiding in Double-Slit Experiments

The physical picture revealed by the interference funnel mechanism, where “**macroscopic matter structures pre-carve geometric guiding potentials in the vacuum,**” is equally applicable to resolving the mystery of non-local guidance in electron double-slit experiments:

- **Spatial Deployment of the Double-Slit Potential Field:** As a macroscopic collective assembly of atoms, the coherent superposition of microscopic beat frequencies generated by the clock redshifts of the double slit’s binding energies projects a macroscopic spatial Moiré interference pattern into the vacuum region past the exit of the slits.
- **De Broglie Wave Guidance (Pilot-Wave Isomorphism):** When a single electron (a Compton-scale soliton) passes through one of the slits, its macroscopic De Broglie envelope couples with the spatial Moiré interference pattern pre-projected into the vacuum by the physical structure of the double slit.
- **Deterministic Trajectory Selection:** The physical trajectory of the electron is guided by the non-linear fluid dynamics of this spatial potential flow, clustering toward the wave antinodes (the potential energy minima) of the

Moiré fringes. This offers a purely geometric and physical realization of classical double-slit interference and the de Broglie-Bohm pilot-wave mechanism, grounded entirely in local non-linear fluid wave dynamics without relying on quantum non-local mysticism.

5 Coupled-Mode Equations with a Nuclear Potential

In Part I, Section 12.4, we established the coupled-mode equations (CMEs) for the forward envelope F and backward envelope B of a free electron. However, to fully describe physical realities under non-uniform potential fields, such as atomic orbitals, the influence of external electric fields must be directly incorporated into the underlying microscopic coupling equations. This section will demonstrate that by introducing an external potential into the microscopic CMEs and performing an algebraic alignment of spatial scales, the intrinsic scale of the system spontaneously expands from the microscopic Compton wavelength λ_c to the macroscopic Bohr radius a_0 , naturally giving rise to both the non-relativistic Schrödinger equation and its relativistic correction terms.

5.1 Microscopic Variation of Coupled-Mode Equations under an External Potential

To introduce electromagnetic interactions into the underlying classical photon fluid equations, we adopt the equivalent minimal coupling rule. Since the nuclear potential $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$ acts as a scalar pressure in space, it directly modulates the intrinsic refractive index of the vacuum medium, thereby inducing a local modulation of the high-frequency clock.

To rigorously maintain the algebraic consistency of dimensions in three-dimensional space, we upgrade the forward and backward envelopes F and B to **two-component Weyl spinors**, and introduce the Pauli matrix vector $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ to construct the matrix form of spatial partial derivatives.

Let the time evolution operator of the system's spatiotemporal wave function undergo the following substitution:

$$i\hbar\partial_t \rightarrow E_{total} - V(r) \quad (37)$$

where $E_{total} = mc^2 - E_b$ represents the steady-state energy incorporating the adaptive redshift binding energy E_b . We introduce the dimensionless spatial coupling coefficient $\kappa = \frac{mc}{\hbar}$. Substituting the above temporal derivative into the first-order

differential equations in the Weyl representation from Section 12.4, and multiplying the equations by $\hbar$, we obtain the microscopic coupled-mode equations evolving within the nuclear potential field:

$$(mc^2 - E_b)F = -i\hbar c(\boldsymbol{\sigma} \cdot \nabla)F + mc^2 B + V(r)F \quad (38)$$

$$(mc^2 - E_b)B = i\hbar c(\boldsymbol{\sigma} \cdot \nabla)B + mc^2 F + V(r)B \quad (39)$$

To clarify the algebraic structure, we define the local effective kinetic energy term $E_{eff}(r) \equiv mc^2 - E_b - V(r)$. The equations can then be rewritten as:

$$E_{eff}F = -i\hbar c(\boldsymbol{\sigma} \cdot \nabla)F + mc^2 B \quad (40)$$

$$E_{eff}B = i\hbar c(\boldsymbol{\sigma} \cdot \nabla)B + mc^2 F \quad (41)$$

At this point, the forward wave envelope spinor F and the backward wave envelope spinor B are strongly coupled through the locally varying energy barrier $E_{eff}(r)$.

5.2 Non-linear Algebraic Alignment of Spatial Scales

To determine the spatial characteristic scale of the envelope F under the influence of an external non-uniform potential field, we must eliminate the backward envelope B .

First, we isolate the algebraic expression for B directly from (40):

$$mc^2 B = E_{eff}F + i\hbar c(\boldsymbol{\sigma} \cdot \nabla)F \quad (42)$$

Next, we apply the first-order spatial derivative operator $i\hbar c(\boldsymbol{\sigma} \cdot \nabla)$ to both sides of (42):

$$mc^2[i\hbar c(\boldsymbol{\sigma} \cdot \nabla)B] = i\hbar c(\boldsymbol{\sigma} \cdot \nabla)(E_{eff}F) + i\hbar c(\boldsymbol{\sigma} \cdot \nabla)[i\hbar c(\boldsymbol{\sigma} \cdot \nabla)F] \quad (43)$$

Utilizing the Pauli operator product identity $(\boldsymbol{\sigma} \cdot \nabla)^2 F = \nabla^2 F$ (under the slowly varying envelope approximation, the curl term $\nabla \times \nabla F \approx 0$ can be neglected) and the Leibniz rule for spatial differentiation, we obtain:

$$mc^2[i\hbar c(\boldsymbol{\sigma} \cdot \nabla)B] = i\hbar c(\boldsymbol{\sigma} \cdot \nabla E_{eff})F + i\hbar cE_{eff}(\boldsymbol{\sigma} \cdot \nabla)F - \hbar^2 c^2 \nabla^2 F \quad (44)$$

Now, multiplying both sides of (41) by the mass constant mc^2 gives:

$$E_{eff}(mc^2 B) = mc^2[i\hbar c(\boldsymbol{\sigma} \cdot \nabla)B] + (mc^2)^2 F \quad (45)$$

Substituting (42) and (44) into (45), we expand both sides:

$$E_{eff}[E_{eff}F + i\hbar c(\boldsymbol{\sigma} \cdot \nabla)F] = [i\hbar c(\boldsymbol{\sigma} \cdot \nabla E_{eff})F + i\hbar cE_{eff}(\boldsymbol{\sigma} \cdot \nabla)F - \hbar^2 c^2 \nabla^2 F] + m^2 c^4 F \quad (46)$$

Thanks to the exquisite algebraic symmetry of the spinor space, the common cross-derivative term $i\hbar c E_{eff}(\boldsymbol{\sigma} \cdot \nabla)F$ on both sides **strictly cancels**. Rearranging terms yields:

$$E_{eff}^2 F = i\hbar c(\boldsymbol{\sigma} \cdot \nabla E_{eff})F - \hbar^2 c^2 \nabla^2 F + m^2 c^4 F \quad (47)$$

Rearranging terms and dividing both sides by $\hbar^2 c^2$, we arrive at the second-order stationary spatial evolution differential equation for the envelope spinor F :

$$\nabla^2 F - i \frac{\boldsymbol{\sigma} \cdot \nabla E_{eff}}{\hbar c} F = \left[\frac{m^2 c^4 - E_{eff}^2}{\hbar^2 c^2} \right] F \quad (48)$$

5.3 Resonant Dissolution of Compton and Bohr Scales

The bracketed term on the right-hand side of (48) dictates the fundamental spatial variation rate of the envelope. We substitute $E_{eff}(r) = mc^2 - (E_b + V)$ into the brackets and expand:

$$m^2 c^4 - E_{eff}^2 = m^2 c^4 - [mc^2 - (E_b + V)]^2 \quad (49)$$

$$= m^2 c^4 - [m^2 c^4 - 2mc^2(E_b + V) + (E_b + V)^2] \quad (50)$$

In the weak-field limit (where the external nuclear potential satisfies $E_b, V \ll mc^2$), we can reasonably neglect the minute quadratic term $(E_b + V)^2$. Consequently, the massive constant term $m^2 c^4$, which represents the microscopic rigid self-trapping limit, is **strictly canceled**.

The remaining term inside the brackets simplifies to:

$$m^2 c^4 - E_{eff}^2 \approx 2mc^2(E_b + V) \quad (51)$$

Substituting this approximation back into (48), and noting that $\nabla E_{eff} = -\nabla V$, we move the energy terms back to the left-hand side to obtain the final equation:

$$\nabla^2 F + i \frac{\boldsymbol{\sigma} \cdot \nabla V}{\hbar c} F - \frac{2m}{\hbar^2} \left(E_b - \frac{e^2}{4\pi\epsilon_0 r} \right) F = 0 \quad (52)$$

5.4 Ontological Reconstruction of the Physical Picture and Insights

By directly incorporating the nuclear potential into the underlying non-linear coupled-mode equations (CMEs), we have not only rigorously derived the non-relativistic Schrödinger equation and its relativistic corrections, but have also demystified and ontologically reconstructed many abstract postulates of standard quantum mechanics (QM). Below is a systematic summary of the core physical insights:

5.4.1 The Forward-Backward Wave Nature of Chiral Weyl Spinors and Positive-Definite Energy

In standard QM, the bispinor $\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$ is typically defined as an abstract complex probability amplitude satisfying the representations of the Poincaré group. However, in the physical Ansatz of the coupled-mode equations (Eq. 52 in Part I), these two components possess highly intuitive classical wave correspondences:

- **Forward Envelope F** : Represents the forward-propagating photon wave packet traveling in the $+z$ direction within the vacuum Bragg grating.
- **Backward Envelope B** : Represents the backward-propagating photon wave packet traveling in the $-z$ direction within the vacuum Bragg grating.

On the level of physical reality, both F and B are ordinary, high-frequency, positive-energy waves. This directly resolves the long-standing “negative-energy paradox” of relativistic quantum mechanics.

In classical field theory, the total energy of a physical system is determined by the T^{00} component of the energy-momentum tensor (the energy density u). In our model, the physical energy density u always manifests as a positive-definite quadratic form:

$$u \propto |A_+|^2 = |F|^2 + |B|^2 \geq 0 \quad (53)$$

Whether representing matter or antimatter states, the physical energy density is strictly positive. This implies that **neither the forward nor the backward waves carry negative energy; they are simply ordinary energy flows propagating in opposite directions in space.**

5.4.2 Phase-Locked Eigenstates of Antimatter

In the Weyl representation, the two self-locked eigenstates of a particle at rest ($p = 0$) are determined by the **relative phase** between the forward and backward waves:

- **Matter State (Electron, $E = +mc^2$)**: The forward and backward waves are locked **in-phase** ($F = B$). Here, the forward and backward energy flows undergo symmetric interference, forming a stable, in-phase stationary standing wave with an energy density of $u \propto 2|F|^2 > 0$.
- **Antimatter State (Positron, $E = -mc^2$)**: The forward and backward waves are locked **out-of-phase** ($F = -B$), meaning their relative phase differs by π . Here, they form an anti-symmetric standing wave; due to the

modulus squaring, the energy density remains positive-definite: $u \propto 2|-F|^2 = 2|F|^2 > 0$.

Given that F and B are ordinary positive-energy waves, how does “antimatter (the positron)” emerge at the fundamental level? Why does an **internal phase-locking relationship that is π out-of-phase** give rise to antimatter characteristics? Is there a deeper physical relationship not captured by these equations? Guided by the previously defined beat delays of positive and negative momentum, we will expand on antimatter in the subsequent section.

5.4.3 Fluid Physical Interpretation of “Large and Small Components” in the Standard Dirac Representation

In standard quantum mechanics, the Dirac representation is commonly used to describe low-speed electrons, where the spinor is split into a large component ϕ and a small component χ . We can project the forward and backward waves (F and B) from the Weyl representation to the Dirac representation using a strict unitary matrix transformation:

$$\Psi_D = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix} \begin{pmatrix} F \\ B \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} F+B \\ F-B \end{pmatrix} \doteq \begin{pmatrix} \phi \\ \chi \end{pmatrix} \quad (54)$$

This transformation provides the most intuitive fluid-dynamical definition of the large and small components:

- **Large Component** $\phi \propto (F+B)$: Physically corresponds to the **in-phase superposition (sum)** of the forward and backward waves, representing the main body of energy of the stationary standing wave.
- **Small Component** $\chi \propto (F-B)$: Physically corresponds to the **out-of-phase residual (difference)** of the forward and backward waves, representing the asymmetry of the macroscopic net momentum flow in space.

When the electron is absolutely at rest ($p = 0 \implies F = B$), the large component $\phi = 2F$ is maximized, while the small component $\chi = 0$ is strictly zero. When the electron moves at low speed ($p \ll mc$), the forward wave must carry slightly more energy than the backward wave to generate macroscopic displacement, resulting in a tiny residual δ ($F \approx B + \delta$). Here, the large component $\phi \approx 2B$ remains high, while the small component $\chi = \delta \propto p$ emerges.

The abstract mathematics in relativistic quantum mechanics stating that “the small component is proportional to momentum p ” thus receives a clear physical explanation: **the small component is not a non-local mixture of antimatter,**

but rather the energy asymmetry residual ($F - B$) that must be introduced between the forward and backward waves to allow the stationary standing wave envelope to undergo macroscopic translation in space.

5.4.4 Phase Resonance and Envelope Relaxation in the Non-Relativistic Limit

Our derivation reveals that the emergence of the non-relativistic Schrödinger equation from the second-order spatiotemporal wave equation algebraically hinges on the cancellation of the $m^2c^4 - E_{eff}^2$ term. Physically, this corresponds to **phase-matching resonance**.

A free electron sustains its self-focusing within the microscopic Compton scale λ_c through its high-frequency intrinsic oscillation (m^2c^4). When an external inhomogeneous scalar potential $V(r)$ permeates the system, it distorts the intrinsic refractive index of the vacuum across space. To prevent dissipative collapse, the system undergoes an adaptive redshift $\Delta\omega$ (binding energy), which perfectly cancels the original self-trapping rigid tension across all spacetime.

This cancellation instantly “releases” the internal tension that previously bound the energy tightly within λ_c . The wave packet undergoes spatial relaxation, expanding and dissolving outward like melting ice. The system only re-establishes a dynamic weak-field equilibrium once it diffuses to a macroscopic boundary dominated by the Coulomb potential gradient—the **Bohr radius** a_0 .

5.4.5 Spin-Orbit Coupling: The Spin Hall Effect of Light in a Gradient-Index Vacuum

The first-order imaginary spatial derivative $+i\frac{\sigma \cdot \nabla V}{\hbar c}F$, which automatically emerges in (52), is fundamentally a unique property of the chiral spinor equation in the Weyl representation.

The spin-1/2 attribute physically corresponds to the **circular polarization** of the underlying photon flow in its self-locked state. In classical wave optics, when a circularly polarized beam propagates through a **Gradient-Index (GRIN) Medium** constructed by the scalar potential $V(r)$, its beam centroid spontaneously shifts transversely due to asymmetric boundary momentum and Berry phase modulation—a phenomenon known in modern optics as the “**Spin Hall Effect of Light (SHEL)**”.

Therefore, spin-orbit coupling gains a unified classical optical explanation: **it is not an abstract dynamical force arising from coordinate transformations, but the real-space mathematical projection of the Spin Hall Effect of Light spontaneously generated when a circularly polarized, stationary matter wave moves through the gradient-index vacuum.**

In macroscopic observations, the classical Schrödinger wave function we measure corresponds to the Dirac large component $\phi = F + B$. When we project the chiral component equations onto the large component representation and retain corrections up to the second order (order $1/m^2$), this first-order imaginary term will naturally transform into the standard Pauli spin-orbit coupling term ($\boldsymbol{\sigma} \cdot (\nabla V \times \mathbf{p})$) and the Darwin term ($\nabla^2 V$) through the non-trivial commutation relation between the spatial momentum operator $\mathbf{p} = -i\hbar\nabla$ and the potential $V(r)$.

Thus, $+i\frac{\boldsymbol{\sigma} \cdot \nabla V}{\hbar c}F$ is not an abstract dynamical postulate, but rather the **unified precursor algebraic source of both spin-orbit coupling and the Darwin term under the microscopic chiral representation**.

5.4.6 The Lattice Rupture Picture of Vacuum Breakdown in Super-heavy Nuclei

When the atomic number $Z \approx 137$ (the limit of the inverse fine-structure constant α^{-1}), the spatial gradient of the external nuclear potential as $r \rightarrow 0$ becomes extremely steep, leading to:

$$\kappa^2 - k_{eff}^2 < 0 \quad (55)$$

Mathematically, the coefficient on the right-hand side of (48) changes sign, transforming the spatial solution from a decaying real exponential (bound state) to an oscillating complex exponential (radiative escape state).

Physically, this represents a state where the spatial shear force of the external nuclear electric field completely overcomes the physical limit of the “self-coherent Bragg reflector (m^2c^4)” that holds the electron together via non-linear self-focusing. The self-locked standing wave of the electron is physically **ruptured** by the external potential, causing the localized electromagnetic energy flow to instantly disassemble and escape outward. This explains why conventional atomic orbital configurations cannot stably exist beyond this limit.

6 From Nonlinear Equations to Antimatter

Having established the self-locking standing wave model of the electron, we must address a more fundamental symmetry question: within the same underlying non-linear, non-local field equations, why do symmetric antimatter (positron) solutions of opposite charge and symmetric physical properties simultaneously exist?

This section will demonstrate that **the physical essence of antimatter’s opposite charge is not an artificially assigned algebraic label, but a direct consequence of the differing phase update relations between the b_0 and b_1 layers carrying momentum within the underlying grid. This temporal**

inversion acts as the cause, rigidly requiring the system to spontaneously generate symmetric resonant solutions with a π phase difference (out-of-phase self-locking) under the same non-linear field equations.

6.1 Cross-Layer Latency and the Physical Definition of the Complex Four-Potential

According to the formulation in Section 1.1 of Part I, the underlying computational grid is partitioned into two layers, b_0 and b_1 . The cross-layer beat delay relationship defines the sign of the momentum. We introduce a complex four-potential $A^\mu(\mathbf{x}, t)$ to describe the physical state of this bilayer grid:

$$A^\mu(\mathbf{x}, t) = A_{b_0}^\mu(\mathbf{x}, t) + iA_{b_1}^\mu(\mathbf{x}, t) \quad (56)$$

where the imaginary unit i plays a critical role: **it is not merely an algebraic symbol, but a “cross-layer information copy latency operator” representing a $\pi/2$ temporal phase shift between the b_0 and b_1 layers.**

For wave packets oscillating at high frequencies along the time axis, this inter-layer latency relationship can be quantitatively described via trigonometric combinations:

1. Electron Sequence ($b_1 \rightarrow b_0$ copy latency):

Information first appears on the b_1 layer and is subsequently copied to the b_0 layer, meaning b_0 lags b_1 . If we set the b_1 layer signal as $\cos(\omega t)$, then the b_0 layer signal is $\cos(\omega t - \pi/2) = \sin(\omega t)$. Mathematically, this corresponds to the standard rotating phase:

$$A_{elec}^\mu \propto \sin(\omega t) + i \cos(\omega t) = i (\cos(\omega t) - i \sin(\omega t)) = i e^{-i\omega t} \quad (57)$$

2. Positron Sequence ($b_0 \rightarrow b_1$ copy latency):

Information first appears on the b_0 layer and is subsequently copied to the b_1 layer, meaning b_1 lags b_0 . If we set the b_0 layer signal as $\cos(\omega t)$, then the b_1 layer signal is $\cos(\omega t - \pi/2) = \sin(\omega t)$. This temporal inversion mathematically corresponds to the counter-rotating complex phase:

$$A_{posi}^\mu \propto \cos(\omega t) + i \sin(\omega t) = e^{i\omega t} \quad (58)$$

It is easy to see that since $(A_{elec}^\mu)^* = \sin(\omega t) - i \cos(\omega t) = -i e^{i\omega t} \propto A_{posi}^\mu$, utilizing the complex conjugation transformation $i \rightarrow -i$ (which physically corresponds to the grid cross-layer temporal reversal), we can strictly map the phase relationship of matter and antimatter (arising from the inversion of the underlying temporal sequence) to a conjugate symmetry of the complex potential field in phase space.

6.2 Classified Definitions of Matter/Antimatter Potentials, Momentum, and Photons

To clarify the physical picture, we provide the following systematic definitions within the bilayer grid framework:

1. Positive/Negative Momentum:

- **Negative Momentum** (p^-): Characterizes the grid shear pressure established by the asymmetric temporal flow from the $b_1 \rightarrow b_0$ layers.
- **Positive Momentum** (p^+): Characterizes the grid shear pressure established by the asymmetric temporal flow from the $b_0 \rightarrow b_1$ layers.

2. Positive/Negative Photons:

- **Negative Photon** (γ^-): High-frequency propagation of entropy momentum density fluctuations on the grid, whose internal phase is locked to the $b_1 \rightarrow b_0$ temporal sequence, characterizing a negative momentum flow.
- **Positive Photon** (γ^+): High-frequency propagation of entropy momentum density fluctuations on the grid, whose internal phase is locked to the $b_0 \rightarrow b_1$ temporal sequence, characterizing a positive momentum flow.

3. Matter/Antimatter Potentials:

- **Matter Potential** A_{elec}^μ : A self-focusing localized potential well formed by the interference of negative photons in an in-phase self-locked state.
- **Antimatter Potential** A_{posi}^μ : A self-focusing localized potential well formed by the interference of positive photons in an out-of-phase self-locked state.

6.3 The Complex Potential and the C-Symmetry of the Field Equation

In non-linear wave physics, both matter and antimatter potentials must be governed by the **same** underlying non-linear, non-local field equation. In an absolute vacuum, the complex four-potential A^μ satisfies:

$$e^{a^2 \square} A^\mu = [1 + g(A_\nu A^{\nu*})] A^\mu \quad (59)$$

We verify the invariance of this non-linear equation under the complex conjugation transformation:

1. **Real Operator:** Since the four-dimensional D'Alembertian operator $\square = \frac{1}{c^2}\partial_t^2 - \nabla^2$ is a pure real differential operator, its non-local exponential operator $e^{a^2\square}$ has purely real coefficients. Thus, for any complex potential field:

$$\left(e^{a^2\square}A^\mu\right)^* = e^{a^2\square}A^{\mu*} \quad (60)$$

2. **Real Non-linearity:** The argument of the grid saturation function g is the field's modulus square $A_\nu A^{\nu*}$. Since the modulus square is strictly real, the non-linear feedback coefficient $[1 + g(A_\nu A^{\nu*})]$ is also purely real:

$$[1 + g(A_\nu A^{\nu*})]^* = 1 + g(A_\nu^* A^\nu) \quad (61)$$

Applying the complex conjugation transformation (which physically corresponds to the grid cross-layer temporal reversal $i \rightarrow -i$) to both sides of the field equation (59) yields:

$$e^{a^2\square}A^{\mu*} = [1 + g(A_\nu^* A^\nu)] A^{\mu*} \quad (62)$$

Conclusion:

If the complex field A^μ is a solution to (59), then its temporally reversed conjugate field $A^{\mu*}$ must also be a strict solution to the equation. This establishes Charge Conjugation Symmetry (C-symmetry) at the mathematical foundation, allowing two perfectly symmetric physical solutions to coexist under the same set of rules.

6.4 Algebraic Emergence of the Positron Trial Solution (Out-of-Phase Locking) in Spinor Space

We now mathematically demonstrate, via the charge conjugation of the Weyl spinor, why the forward and backward wave packets of a positron must spontaneously generate a relative phase difference of π to achieve standing wave resonance under a reversed temporal sequence.

In Section 12.1, we constructed the two-component Weyl spinor Ψ of a matter wave using the forward envelope F and backward envelope B (omitting the common carrier phase):

$$\Psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} \doteq \begin{pmatrix} F \\ B \end{pmatrix} \quad (63)$$

In relativistic quantum mechanics, the charge conjugation operator $\hat{C}$ corresponding to the underlying $i \rightarrow -i$ conjugation is defined as:

$$\Psi^c = \hat{C}\Psi = -i\sigma_y\Psi^* \quad (64)$$

where $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. Expanding this operator yields:

$$-i\sigma_y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (65)$$

Substituting the matter wave spinor Ψ to perform the conjugation calculation (for the static ground state, the envelope functions F and B can be taken as real):

$$\Psi^c = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F^* \\ B^* \end{pmatrix} = \begin{pmatrix} -B \\ F \end{pmatrix} \quad (66)$$

Now, we compare the intrinsic resonant states of matter and antimatter:

1. Electron (In-phase self-locking, matter eigenstate):

To maintain constructive interference in the standard temporal flow (i), its trial solution requires the forward and backward waves to remain in-phase:

$$\Psi_{elec} = \begin{pmatrix} F \\ F \end{pmatrix} \quad (\text{i.e., } F = B) \quad (67)$$

2. Positron (Out-of-phase self-locking, antimatter eigenstate):

Because the underlying phase relationship is inverted ($i \rightarrow -i$), the system is forcibly projected onto the conjugate branch Ψ^c . Substituting the in-phase resonance trial solution of the electron into the charge conjugation relation (66) directly yields the intrinsic resonant state of the positron, Ψ_{posi} :

$$\Psi_{posi} = \Psi_{elec}^c = \begin{pmatrix} -F \\ F \end{pmatrix} \quad (68)$$

Letting the forward and backward envelope components of the positron be F' and B' respectively, such that $\Psi_{posi} = \begin{pmatrix} F' \\ B' \end{pmatrix}$, comparison yields:

$$F' = -F, \quad B' = F \implies F' = -B' \quad (\text{i.e., } F' = e^{i\pi} B') \quad (69)$$

Conclusion:

A phase difference of π between the forward and backward wave packets is the unique, self-consistent geometric response generated by the positron to satisfy the same non-linear field equation within an inverted grid temporal sequence. This “out-of-phase self-locking” ensures that antimatter can similarly form stable, self-focusing standing waves in phase space without dissolving.

7 The Fluid Topological Origin of Planck’s Constant and the Vacuum Restoration Mechanism

Having concluded the discussions above, we must return to reflect on the cornerstone of all quantum phenomena in traditional quantum theory—the Planck energy-frequency relation, $E = h\nu$. While serving as the bedrock of quantum mechanics, this relation itself is treated as an a priori axiom requiring no explanation.

If a photon is fundamentally a macroscopic fluid topological soliton (an envelope standing wave) on the vacuum computing grid, why is its total energy strictly proportional to its frequency, and completely independent of the absolute length L of its wave train?

This section will demonstrate, through two parallel perspectives of pure classical non-linear physics—“Energy Compensation of Logarithmic Dispersion” and “Vacuum Restoration under the Soliton Area Theorem”—that Planck’s constant h is not a mysterious quantum step, but rather the **absolute geometric action threshold** that the vacuum grid, acting as an elastic medium, must preserve to maintain the stable propagation of a fundamental topological soliton (the photon).

7.1 Perspective I: D’Alembert Spatiotemporal Cancellation, Logarithmic Dispersion, and the Algebraic Closed Loop of Energy Density

A single photon in the physical world is not an idealized, infinitely long plane wave, but a wave packet with a finite coherence length (the wave train length L). Its mathematical morphology consists of a high-frequency “elastic carrier wave” and a surrounding “plastic envelope.”

1. Exact Spatiotemporal Cancellation and the Rupture of Finite Wave Trains

For an infinitely long, idealized coherent wave train ($L \rightarrow \infty$), when the four-dimensional spatiotemporal D’Alembertian operator $\square = \frac{1}{c^2}\partial_t^2 - \nabla^2$ acts upon it, the temporal second-order curvature ($-k^2$) and the spatial propagation curvature ($+k^2$) achieve a **perfect spatiotemporal cancellation** ($-k^2 + k^2 \equiv 0$). At this point, the wave train is entirely dispersion-free and can propagate freely through the vacuum without any nonlinear confinement.

However, any physical wave packet possesses a finite length L . At the head and tail of the wave train (the end-point regions), the envelope must smoothly climb from absolute vacuum (0) to a maximum value, and then fall back to 0.

The presence of this envelope gradient (∇F) **breaks the exact spatiotemporal cancellation of the D'Alembertian operator.**

2. Logarithmic Dispersion Destructive Force of Topological Defects

Within the “computational grid” framework of this theory, due to the complex internal momentum interactions of the soliton under non-local nonlinear constraints, the group velocity dispersion (GVD) tearing force F_{disp} accumulated by the D'Alembertian cancellation failure does not follow a simple power law, but rather obeys a **logarithmic scaling law** driven by grid recursion:

$$F_{disp} \propto \ln\left(\frac{1}{L}\right) \quad (70)$$

As the wave train shortens ($L \rightarrow 0$), the rupture of spatiotemporal cancellation becomes more severe, causing the tearing force to diverge logarithmically; conversely, as the wave train lengthens ($L \rightarrow \infty$), the spatiotemporal cancellation approaches perfection, and the tearing force decays logarithmically to zero.

3. Logarithmic Nonlinear Confinement and the Balance of Force

To prevent the wave packet from being torn apart by the end-point dispersion, the vacuum grid must provide a nonlinear self-focusing attractive force F_{NL} to counter the dispersion.

Under our framework, the **saturated nonlinear feedback function $g(u)$ of the vacuum is logarithmic.** Meaning the nonlinear confinement force strictly satisfies:

$$F_{NL} \propto \ln(u) \quad (71)$$

(where u is the local grid energy density).

To maintain the stable existence of the soliton, the nonlinear confinement force must strictly balance the dispersion destructive force ($F_{NL} = F_{disp}$), yielding the algebraic relationship:

$$\ln(u) \propto \ln\left(\frac{1}{L}\right) \quad (72)$$

Raising both sides to exponential powers yields the geometric requirement for the soliton's energy density:

$$u \propto \frac{1}{L} \quad (73)$$

Geometric Conclusion: In a logarithmic nonlinear fluid, to counter logarithmic dispersion, the local energy density must scale inversely with the wave packet length. Consequently, its integral over the spatial volume—namely, the photon's **total energy** $E = \int u dV \propto u \times L$ —perfectly eliminates L under any stretching

or compression of the wave train, remaining **strictly constant**. A photon of 1000 wavelengths and a photon of 10 wavelengths possess the exact same independent total energy E , representing a topologically indivisible, closed unit.

7.2 Perspective II: The Soliton Area Theorem and Spatiotemporal Vacuum Restoration

If the total energy E is independent of the wave train’s length, why must it be strictly proportional to the frequency ν (or angular frequency ω)? To answer this, we must introduce the profound concept of “**Vacuum Restoration**” from non-linear soliton physics.

1. Demanding Condition for Lossless Propagation: Spatiotemporal Dual Restoration

As a fundamental particle, a photon can travel tens of billions of light-years in a vacuum without dissipating. This implies that after passing through an infinitesimal region of the vacuum grid, the photon must not leave any “aftershocks” or “radiation wakes.” The photon wave train must not only physically exit the region in **space**, but must also require the grid’s **temporal state** and **potential field deformation** to return precisely to their ground states. This is termed “spatiotemporal dual restoration.”

2. The Soliton Area Theorem

In classical non-linear optics (such as the McCall-Hahn model of self-induced transparency), for a pulse to achieve perfect, wake-free restoration in a medium, its envelope must satisfy a strict **Area Theorem**: the integral of the “action angle” exerted by the pulse on the medium from start to finish must be an intrinsic topological constant θ_c (e.g., 2π).

On the underlying four-dimensional computational grid, to distort the vacuum grid from its ground state and perfectly twist it back (forming a closed soliton knot with a topological charge of 1), the spatiotemporal evolution integral of the four-dimensional electromagnetic potential must reach a fixed geometric threshold.

3. Conservation of Action and the Emergence of $E = h\nu$

In fluid dynamics, the cost paid by the system to complete such a full topological inversion is not energy, but **action**.

According to the grid energy density defined in this theory, $u \propto \omega^2 |\Psi|^2$. When a complete soliton passes through the grid, the “wave action S ” (the spatial integral of

its generalized momentum, $\propto \omega|\Psi|^2$) must equal the geometric constant S_0 allowed for the grid to maintain a minimal topological knot:

$$S_{total} = \int \frac{u}{\omega} dV = S_0 \quad (\text{the intrinsic topological invariant of the vacuum grid}) \quad (74)$$

Since the frequency ω of a single coherent wave train is uniform across the entire domain, we can pull it out of the integral:

$$\frac{1}{\omega} \int u dV = S_0 \quad (75)$$

which yields:

$$\frac{E}{\omega} = S_0 \implies E = S_0 \cdot \omega \quad (76)$$

Topological Conclusion: If we name this intrinsic minimal topological closure constant S_0 of the vacuum grid (after scaling) the **Planck constant** $\hbar$, we naturally obtain $E = \hbar\omega = h\nu$.

7.3 Conclusion: The Topological Physical Significance of Planck's Constant

Combining the two perspectives, we extract the real physical picture of Planck's constant from pure fluid dynamics:

Due to the rupture of the D'Alembertian spatiotemporal cancellation, a finite wave packet inevitably faces dispersive tearing. The frequency ν determines the “severity” with which the wave train carrier wave distorts spacetime; the higher the frequency, the more violently the end-point dispersion attempts to rip the soliton apart. To counter this destructive force that rises with frequency, and to achieve the “perfect restoration of the spatiotemporal vacuum (leaving no dissipative wake),” the wave packet must pay a corresponding amount of energy E as “stabilizing collateral” to provide the necessary logarithmic non-linear self-focusing pull.

From this perspective, the essence of Planck's constant h is by no means an arbitrarily assumed, inexplicable energy discontinuity step, but rather the “minimum wave action conservation threshold” that the vacuum's four-dimensional computational medium must maintain to form and preserve a minimal closed topological vortex soliton (the photon). We argue that beneath this conceptual proof, there must exist deeper computational reasons constrained by grid rules.